

usage, contributing to a 25% increase in adoption compared to cloud-dependent alternatives.

SLM Smart Assistants: Voice commands in wearables without connectivity

SLM-powered smart assistants have redefined wearable technology by enabling offline voice command processing. Fitness trackers and smartwatches can now execute tasks like setting reminders, tracking activities, and controlling music playback without relying on cloud-based services. These localised capabilities ensure faster response times and greater privacy, as sensitive voice data remains on the device. The integration of SLMs has also reduced the computational burden by 30%, extending battery life for wearables by an average of 20 hours per charge. This advancement is particularly significant as wearable shipments surpassed 250 million units globally in 2024.

PocketLens AI: Summarizing printed books via smartphone camera

PocketLens AI utilises lightweight models to analyse and summarise content captured through a smartphone's camera. Aimed at students and professionals, this application simplifies research by extracting key points from textbooks and documents. With support for over 30 languages and specialised features for STEM texts, the app has been downloaded over 10 million times since its launch in early 2024. Its ability to integrate seamlessly with note-taking apps and digital libraries makes it a valuable tool for enhancing productivity.



Instant Transcriptions: Fast speech-to-text conversions with LiveScribe

SLM-driven tools like LiveScribe have made speech-to-text transcription faster and more accessible. These tools convert spoken words into text in real time, catering to journalists, content creators, and corporate professionals. LiveScribe's 2024 update introduced context-aware tagging and multilingual transcription support for 15 additional languages, increasing its user base by 40%. With a transcription accuracy rate of 96% in offline mode,

LiveScribe has become an indispensable tool for users working in connectivity-challenged environments.

IoT Ecosystems: Local voice commands for appliance control

SLMs play a critical role in expanding the functionality of IoT ecosystems. They enable users to control multiple smart appliances locally via voice commands, eliminating the need for cloud-based processing. For instance, users can manage lighting, thermostats, and security cameras using a centralised hub powered by an SLM. This architecture reduces latency by 25%, enhances security by keeping data localised, and ensures uninterrupted service even during network outages. In 2024, the global smart home market grew by 18%, with SLM-driven IoT solutions accounting for a significant share of new deployments.

Energy Efficiency

SLMs are driving significant advancements in energy-efficient AI applications. Their streamlined architectures reduce the computational load, making them suitable for devices and systems prioritising sustainability. Here are some of the key developments in 2024:

GreenBERT: Reduced energy consumption in enterprise data centres

GreenBERT is an SLM variant designed specifically for data centers, optimising natural language processing tasks with minimal energy use. In 2024, it achieved a 30% reduction in power consumption compared to traditional models, saving enterprises an estimated \$1 billion annually in energy costs. By employing quantisation techniques and hardware-efficient designs, GreenBERT has become a cornerstone for businesses aiming to lower operational expenses while meeting stringent environmental regulations.

Tesla PowerWall AI: Energy storage and distribution optimisation

Tesla's PowerWall AI incorporates SLMs to optimise energy storage and distribution in smart homes. By analysing energy consumption patterns, solar panel output, and local electricity tariffs, the system recommends cost-effective usage strategies. Tesla reported a 20% increase in energy savings for PowerWall users in 2024, reflecting the system's ability to adapt to real-time changes in energy supply and demand. This innovation aligns with the broader push for sustainable living, with over 500,000 units sold this year alone.

Low-Energy AI Sensors: Crop monitoring in remote fields

SLM-powered sensors are revolutionising agriculture by enabling real-time crop monitoring in remote locations. These sensors analyse soil moisture, temperature, and weather conditions to provide